

Part: A

1 ☐ Compare and distinction the Thomson, Rutherford and Bohr atomic models in terms of the following terms—

- Assumptions
- Success
- Limitations

☐ J.J. Thomson Atomic Model

Assumptions:

Thomson proposed that, the atom is a uniform sphere of positive charge in which negatively charged e^- are embedded. The positive charge is continuously distributed throughout the atom and the electrons are arranged within the sphere such that the atoms remain electrically neutral.

Success:

- Introduced the concept of subatomic particles (e^-) into atomic structure
- Successfully explained the electrical neutrality of atoms
- Provided the first internal structural model of the atom
- Served as a foundation for further experimental investigations

Limitations:

- Failed to explain the results of α -particle scattering experiment

- Could not account for the existence of a central nucleus
- Lacked explanation for atomic stability
- Unable to explain discrete atomic spectra

❑ Ernest Rutherford Atomic Model

Assumptions:

Rutherford proposed that, the atom contains a small, dense, positively charged nucleus at its center, which holds most of the atomic mass. Electrons revolve around this nucleus under electrostatic attraction. The majority of the atom consists of empty space. This model is based on classical mechanics and Coulomb's law of electrostatic force

Success:

- Successfully explained α -particle scattering experiment
- Led to the discovery of the atomic nucleus
- Established that atoms are mostly empty space
- Formed the basis for the nuclear model of the atom

Limitations:

- According to the classical physics, orbiting electron should lose energy and collapse into the nucleus
- Failed to explain atomic stability

Part: B

2⁽ⁱ⁾ □ Explain qualitatively how this experiment results contradicts Thomson Model. (α -particle scattering)

In Thomson's atomic model, the atom is assumed to be a uniform sphere of positive charge with e^- embedded inside it. Therefore an incoming α -particle would experience only weak and gradually distributed electrostatic forces, leading to at most small deflection.

However, the experiment of Rutherford showed that while most α -particles pass through the foil, a few are deflected through large angles or even backward scattering. Such strong deflections can only occur if the positive charge is concentrated in a very small, dense nucleus not spread uniformly.

Thus, the observation of large-angle scattering directly contradicts Thomson's model.

2⁽ⁱⁱ⁾ □ Derive the expression for the closest distance of approach for a 5 MeV α -particle.

At closest distance, velocity becomes zero

Now,

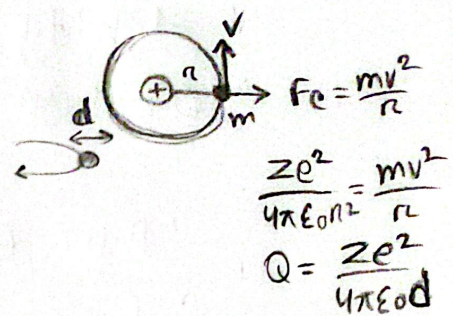
$$KE = PE$$

$$\Rightarrow KE = \frac{1}{4\pi\epsilon_0} \cdot \frac{2eZe}{d}$$

$$\Rightarrow d = \frac{1}{4\pi\epsilon_0} \cdot \frac{2Ze^2}{KE}$$

$$= 9 \times 10^9 \times \frac{2 \times 79 \times (1.6 \times 10^{-19})^2}{5 \times 1.6 \times 10^{-13}}$$

$$= 4.55 \times 10^{-14} \text{ m}$$



$\therefore$ The closest distance is $4.55 \times 10^{-14} \text{ m}$ Ans.

Part: 2

3 ☐ Analytical observation explaining how and why each successive model emerged from the shortcomings of its predecessor

The atomic models evolved as a direct response to the limitations of earlier theories

The model of Thomson described the atom as a uniform positive sphere with embedded electrons. It explained electrical neutrality but failed to account for large angle scattering observed in experiments

This was then corrected by Rutherford, who proposed a dense central nucleus with e^- revolving around it. This explained scattering results but couldn't justify atomic stability or discrete line spectra

To resolve these issues, Bohr introduced quantized orbits with fixed energy levels, successfully explaining

stability and hydrogen spectra, though limited to single electron atoms

Thus, each model emerged to overcome the specific failures of its predecessor, progressively improving the understanding of atomic structure

Thomson's Model $\rightarrow$ Rutherford's Model
Bohr's Model $\leftarrow$

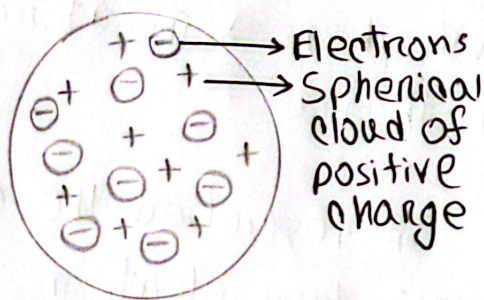


Fig: Thomson's Model

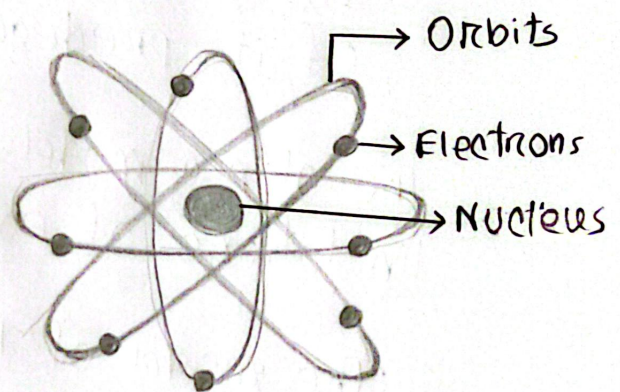


Fig: Rutherford's Model

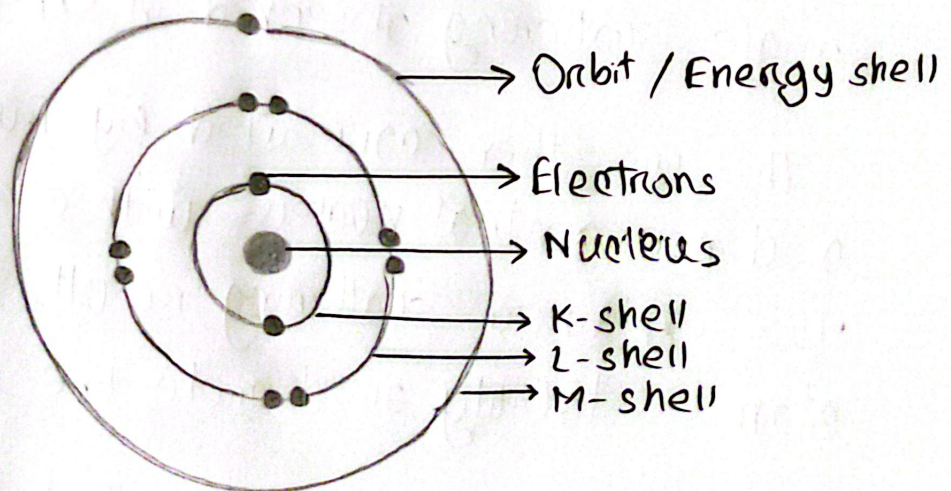


Fig: Bohr's Model